Name & Surname **Polimi ID**

Exercise 1

A. Please describe and show an example of Parallel Data Management Patterns: pipeline. (1)

B. What dependencies can be eliminated by renaming the variables? (1)

C. Where and under what conditions can the following code be parallelized? Is there a way to write the code for better parallelization if these conditions are met? Assume that a and b store integers. (2)

```
for (i=3; i<100; i++) {
        for (j=1; j<100; j++)
            a[i][0] *= f(a[i][j]);
        for (j=0; j<100; j++)
            b[i-3][j] += f(b[i][j]);
}
```

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Exercise 2

A. Please describe the parallel Scatter pattern, including assumptions, PRAM-based complexity, and examples. (2)

B. Please discuss how the Pack parallel pattern works. How is it implemented (pseudo code required and PRAM-based complexity)? (2)

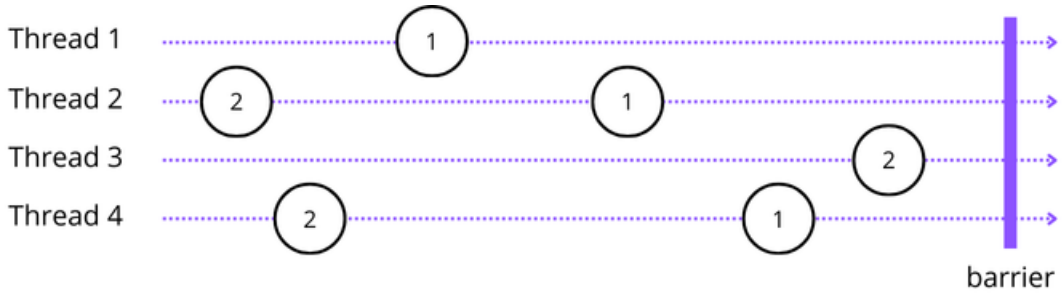
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Exercise 3

A. List similarities and differences between a `single` and a `master` region in OpenMP. (1)

B. What happens when threads encounter a barrier in Pthreads? Is the same behavior achievable through joining? (1) **This question can be skipped if you got 1 or 3 points in the second challenge.**

C. Describe how each thread in the following diagram behaves at runtime as it encounters points marked with 1, representing `pragma omp cancel`, and with 2, representing `pragma omp cancellation point`. (2) **This question can be skipped if you got 2 or 3 points in the second challenge.**



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Exercise 4

A. What do I get from an MPI implementation, e.g. from OpenMPI? (1)

B. Make an **example** where non-blocking semantics does not help with solving an MPI deadlock. (1)

C. How MPI_Reduce work when the send and receive data are buffers with a single element? What happens when the buffers have more than one element? Describe how the result is computed. (2)